DreamAI

Technical & Product Plan

AI-powered nightmare treatment using Image Rehearsal Therapy, dream capture, and personalized video generation to reduce nightmare frequency and support PTSD recovery.

21 min
Military suicide rate
1 every 21 minutes

105%
Increased risk
Frequent nightmares + suicide

7-14d
IRT effectiveness
Nightmare change timeline

$0.46
Cost per cycle
AI image + audio + assembly

Project	DreamAI -- Dream Engineering Platform
Partners	Dr. Michael Breus (Sleep Science) + Cristian Mendivelso (Technical)
Target	Military combat veterans with PTSD nightmares
Phase	MVP -- Functional prototype for validation study
Date	March 2026

01 THE PROBLEM

Nightmares are a modifiable, independent suicide risk factor

Research consistently shows that frequent nightmares are linked to suicidal ideation, plans, attempts, and death -- even after adjusting for depression and other disorders (Bernert 2015, Littlewood 2016). In a Finnish cohort of 36,211 adults, frequent nightmares raised suicide risk by 105% versus none (Tanskanen 2001). Among frontline COVID-19 workers, nightmares fully mediated 66% of the link between trauma exposure and suicidal ideation (Que 2022).

Statistic	Value	Source
Military suicide rate	1 every 21 minutes	DoD 2024
Nightmare-suicide risk increase (frequent)	+105%	Tanskanen 2001
Nightmare-suicide risk increase (occasional)	+57%	Tanskanen 2001
Nightmare mediation of trauma-suicide link	66%	Que 2022
IRT efficacy consensus	84% of studies confirm	Consensus.app
IRT timeline to change nightmare	7-14 days	Krakow 2001
IRT long-term effect duration	4+ years sustained	Sierro 2020

The opportunity

Image Rehearsal Therapy (IRT) is the gold standard for nightmare treatment -- a brief, CBT-based intervention where patients change their nightmare storyline and rehearse the new version daily. It works in 7-14 days and effects last 4+ years. But it requires a therapist, writing exercises, and mental visualization. DreamAI replaces all three with AI-guided voice capture, intelligent rescripting, and personalized video generation -- making IRT accessible to anyone with a phone.

02 THE SOLUTION

DreamAI digitizes the IRT protocol into 6 steps

User Journey -- One Nightmare Cycle (7-14 days)

1 NIGHT CAPTURE
3:00 AM -- 2 min
- Wake from nightmare
- One-tap voice recording
- AI safety screen active
- Raw transcript saved

2 MORNING REFINE
8:00 AM -- 5 min
- AI asks follow-up Qs
- Sensory detail extraction
- Emotional intensity rating
- Dream diary auto-filled

3 AFTERNOON RESCRIPT
2:00 PM -- 5 min
- AI proposes 3 endings
- User picks or modifies
- Full narrative written
- Scene breakdown created

4 VIDEO GENERATION
Background -- 15 min
- 6 scene images generated
- TTS narration recorded
- FFmpeg assembles video
- Push notification sent

5 NIGHTLY REHEARSAL
10:00 PM -- 3 min
- Full-screen immersive
- Earbuds recommended
- Post-watch check-in
- Progress tracked

6 TRACK & ITERATE
Ongoing -- daily
- Nightmare freq. graphed
- Distress trend (0-10)
- AI suggests next steps
- Data ready for study

Clinical protocol mapping

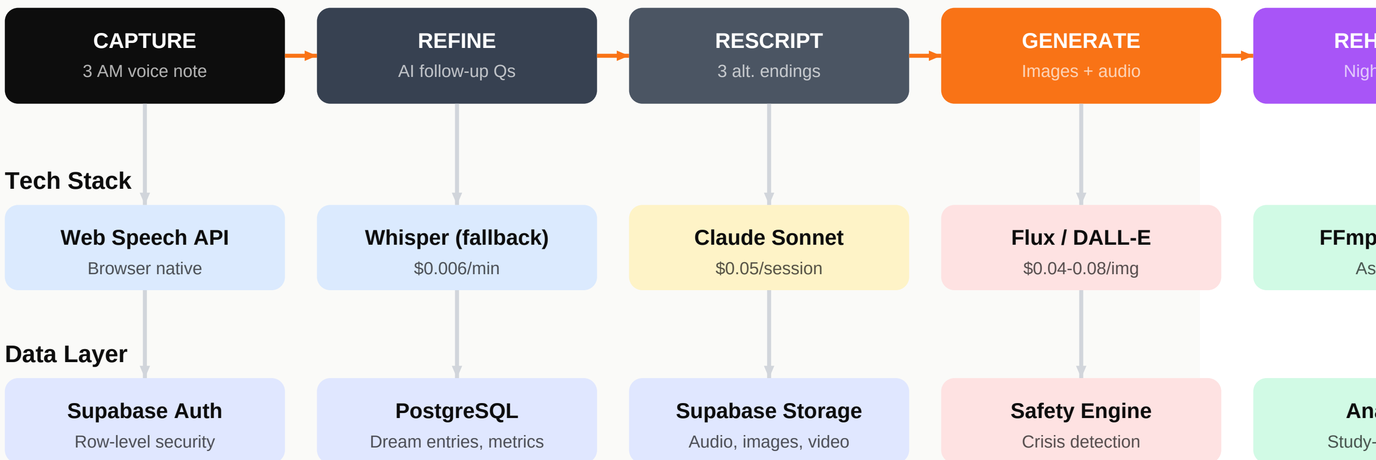
IRT Step (Research)	DreamAI Implementation	Module
1. Psychoeducation	Onboarding flow: 60s video + interactive explainer on how IRT works and why nightmares can change	Onboarding
2. Nightmare selection	User records nightmare via voice immediately after waking; AI identifies the target nightmare	Capture
3. Imagery skills training	Guided pleasant visualization exercise before first rescript (safe-place exercise)	Pre-rescript
4. Nightmare narrative	AI transcribes voice, asks follow-up Qs for sensory detail, auto-fills 9-field dream diary	Refine
5. Imagery rescripting	AI proposes 3 alternative endings (mastery/transformation/safety); user picks or modifies	Rescript
6. Daily rehearsal	AI-generated 2-3 min video with narration + ambient audio; nightly before bed	Video Player

7. Tracking + follow-up	Dashboard: nightmare frequency, distress trend, sleep quality; study-ready data	Analytics
8. Safety layer	Real-time crisis language detection; immediate 988 routing + warm handoff	Safety

03 SYSTEM ARCHITECTURE

System Architecture -- DreamAI MVP

5-phase pipeline: Capture > Refine > Rescript > Generate > Rehearse



COST PER NIGHTMARE CYCLE

~$0.46 Voice $0.01 + AI $0.05 + Images $0.30 + TTS $0.10 = 100 users x 4 cycles/mo = $184/mo

Database Schema (Supabase PostgreSQL)

users	dream_entries	rescripts	rehearsals
id (PK)	id (PK)	id (PK)	id (PK)
email	user_id (FK)	entry_id (FK)	rescript_id (FK)
created_at	recorded_at	ending_type	watched_at
onboarding_done	audio_url	narrative	completion %
sleep_device	transcript	scenes (jsonb)	feeling 0-10
	structured (jsonb)	video_url	
	intensity 0-10		

04 VIDEO GENERATION STRATEGY

Why hybrid (images + motion + audio) beats full AI video

	Full AI Video (Runway/Kling)	Hybrid Approach (Our Choice)	Winner
Cost per video	$9.00 (18 clips)	$0.40 (6 images + TTS)	Hybrid (22x)
Generation time	40-60 minutes	5-8 minutes	Hybrid (8x)
Content control	Unpredictable frames	Full control per scene	Hybrid
Trauma safety	Risk of disturbing AI artifacts	Every frame reviewed	Hybrid
Consistency	Style varies clip to clip	Consistent style via prompt	Hybrid
Immersion	Higher motion realism	Audio narration compensates	Draw

Video assembly pipeline

Scene Decomposition

Claude AI breaks rescripted narrative into 5-6 discrete visual scenes with camera direction, mood, and key elements.

Image Generation (Parallel)

Flux or DALL-E generates one image per scene. Same style prompt prefix ensures consistency. Promise.allSettled with retry.

Narration Generation

OpenAI TTS reads the rescripted narrative in a calm, warm voice. Paced for 2-3 minute total length.

FFmpeg Assembly

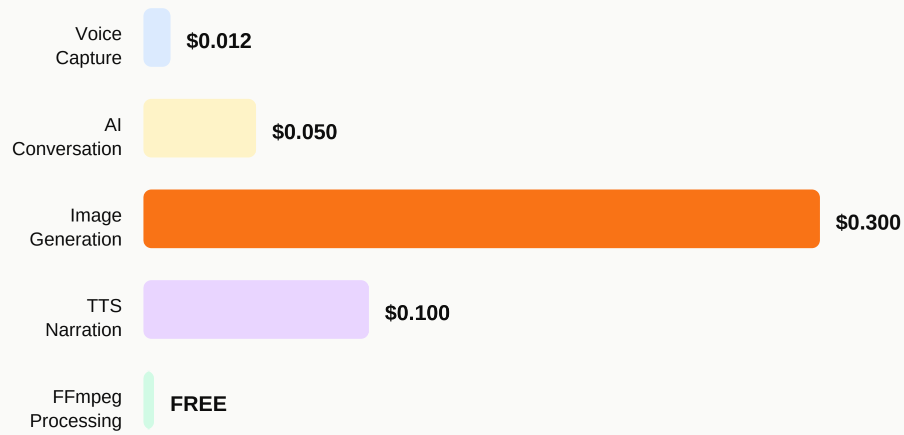
Images get Ken Burns motion (slow zoom/pan). Crossfade transitions (1.5s). Narration + ambient music track layered.

Delivery

MP4 stored in Supabase Storage. Push notification: 'Your DreamAI session is ready.' Full-screen player with earbuds prompt.

05 COST ANALYSIS & TIMELINE

Cost Analysis -- Per Nightmare Cycle



TOTAL PER CYCLE: $0.46 **100 users x 4/mo = $184/mo**

Monthly cost projection at scale

Users	Cycles/mo	API Cost	Infra Cost	Total/mo
10 (pilot)	40	$18	$5	$23
50	200	$92	$10	$102
100 (MVP target)	400	$184	$15	$199
500	2,000	$920	$50	$970
1,000	4,000	$1,840	$100	$1,940

Development Timeline

06 MVP SCOPE & KEY DECISIONS

What we build (and what we don't)

IN SCOPE (MVP)	Priority	OUT OF SCOPE (Future)
Voice-first dream capture (1-tap)	P0	Wearable integration (Oura, Whoop)
AI follow-up conversation	P0	VR/AR version
Safety/crisis detection layer	P0	Lucid dream training module
3-ending AI rescripting	P0	Therapist dashboard
Hybrid video generation	P0	Native iOS/Android app
Immersive rehearsal player	P0	Multi-language support
9-field dream diary (auto-filled)	P1	Social/community features
Progress dashboard + metrics	P1	Wearable sleep stage data
Imagery skills training (onboarding)	P1	Advanced dream engineering (TDI)
Study-ready data collection	P1	Real-time EEG integration

Open questions (awaiting Dr. Breus)

#	Question	Impact on Build	Status
1	Sample nightmare narrative for demo	Needed for Thursday video demo	SENT
2	Crisis/safety protocol specifics	Defines safety layer behavior	SENT
3	Literal vs symbolic video style	Determines image generation prompts	SENT
4	Imagery skills training exercise	Onboarding guided visualization	SENT
5	Writing vs speaking therapeutic value	Voice-only vs transcript review UX	SENT
6	How many rescripts per nightmare	Defines the product loop	SENT
7	Clinical outcome measures for study	Dashboard metrics + data schema	SENT

07 NEXT STEPS

Immediate actions (this week)

MAR 17-18

MONDAY-TUESDAY: Foundation

Set up Next.js project, Supabase database, auth flow. Build voice recording component with Web Speech API. Implement dream diary data model.

MAR 19

WEDNESDAY: AI + Video

Integrate Claude API for follow-up questions and rescripting. Build image generation pipeline with Flux. Set up FFmpeg video assembly on VPS.

MAR 20

THURSDAY: Demo Day

Assemble end-to-end demo. Record walkthrough video. Present to Dr. Breus: working voice capture + at least 1 complete generated video.

MAR 21

FRIDAY: Iterate

Incorporate Breus feedback. Refine video quality. Add safety layer skeleton. Plan Week 2 sprint.

Path to pilot study

Phase	Timeline	Goal	Users
MVP Build	Weeks 1-4	Functional prototype, end-to-end flow working	Internal testing
Alpha Test	Weeks 5-6	5-10 users, Breus's patients or contacts	5-10
Refinement	Weeks 7-8	Fix issues, improve video quality, add tracking	--
Pilot Study	Weeks 9-16	Controlled study: IRT standard vs DreamAI	50-100
Results + Paper	Week 16-20	Analyze data, publish results, patent filing	--
Free Launch (Military)	Week 20+	Partner with VA / military orgs, give away free	1,000+

"We may have the opportunity to change a lot of people's lives, dude. The impact is gonna be awesome."

-- Dr. Michael Breus, March 16, 2026